

A Protocol for Sleep Fragmentation in persons with and without low back pain: Toward Understanding Effects on Pain and Neuroinflammation

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Background and Rationale



Bidirectional relationship

Pain can disrupt sleep, while disturbed sleep may increase pain sensitivity[1-3]



Sleep disturbance in chronic low back pain (CLBP)

Sleep problems are common, up to 53% experience chronic insomnia, associated with greater pain severity and disability[4-5]



Mechanistic gap

The mechanisms through which sleep disturbances influence chronic pain remain insufficiently understood



Study Objective

To determine whether experimentally fragmented sleep alter pain sensitivity and brain mechanisms in persons with CLBP and pain-free controls

Study Design

Randomised cross-over study

Eligibility criteria:

18-60 years old | Non-smoker | No night-shift work

'Good' sleeper: PSQI≤5, ISI≤7, ESS≤10, Clinical sleep screening

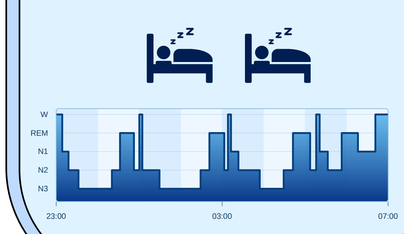
CLBP* (n = 18) | Pain-free controls (n = 18)

*CLBP for at least three months' duration on most days, with a pain rating above 3/10

Randomised condition order

Undisturbed sleep (US) (2 consecutive nights)

Fragmented sleep (FA) (2 consecutive nights)



2-6 weeks washout period

Test day after sleep condition

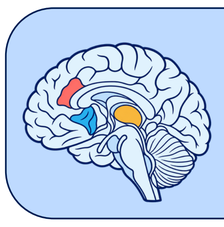
Questionnaires (morning of test day)

Quantitative sensory testing

Magnetic resonance imaging (MRI) scan

Blood sample

Outcome Measures



Brain neurochemistry: ^{1}H-MRS

Regions:
- Pregenual Anterior Cingulate Cortex
- Thalamus
- Anterior Insula

Metabolites: Choline, Myo-inositol*

*Markers associated with neuroinflammatory processes

Brain function

Task-based functional MRI during experimentally evoked back pain

Pain sensitivity

Quantitative sensory testing
- Pressure pain thresholds (calf, forearm, back)
- Temporal summation
- Conditioned pain modulation

Sleep
- Polysomnography (sleep staging)
- Actigraphy
- Sleep diary

Experience sampling method

Daily pain, affect, sleepiness assessment during control week following test day (8 assessment/day)

Questionnaires

Sleep-, pain- and affect-related questionnaires administered at baseline and on the morning of the test day

Hypotheses

We hypothesize sleep fragmentation to..

1. **Brain responses**
   Increase neuroinflammation-related metabolites and pain-evoked brain activity
2. **Pain processing**
   Increase experimental pain sensitivity and impair endogenous pain modulation
3. **Mechanistic coupling**
   Brain changes will be associated with larger increases in pain sensitivity

Greater effects of sleep fragmentation are expected in persons with CLBP than in pain-free controls

Study Progress

Target **N = 36**

Eligible and enrolled - 23 / 36

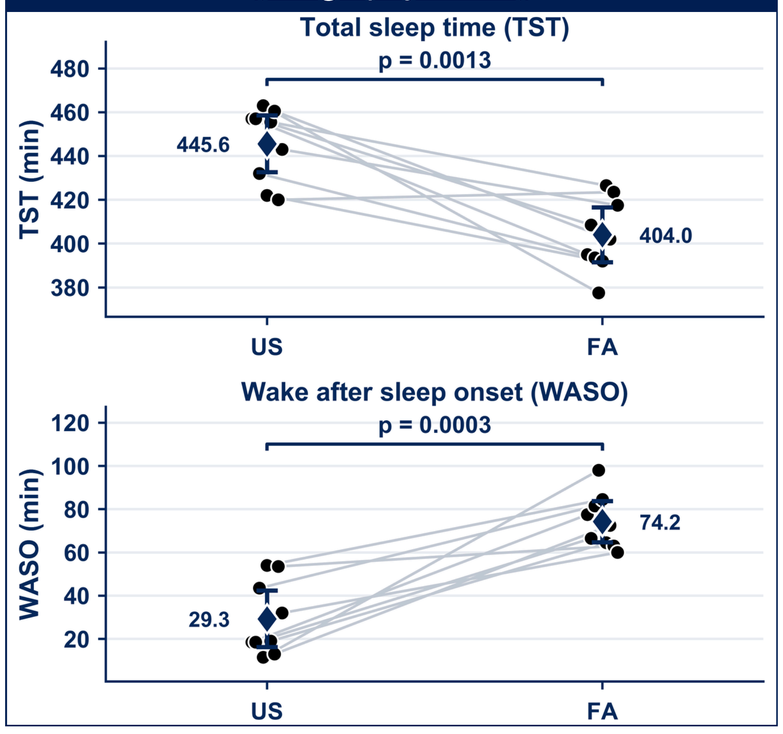
Data collection progress

Eligible and enrolled
8 CLBP (75% Female, age: 34 ±13.4, BMI: 23 ±2.5)
15 CON (67% Female, age: 29 ±8.7, BMI: 23 ±3.3)

Data collection complete
2 CLBP (50% Female, age: 23 ±2.1, BMI: 23 ±2.9)
10 CON (70% Female, age: 29 ±9.8, BMI: 22 ±3.1)

Projected data collection completion
May 2027

Preliminary Protocol Validation (Actigraphy-based)



Take-home Message

By experimentally fragmenting sleep, this study tests whether disrupted sleep alters **pain sensitivity and pain-related brain mechanisms**, and whether **these effects differ between persons with CLBP and pain-free controls.**

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